

CY 670: Fluorescence spectroscopy, microscopy and applications

Course Code	CY 670
Course Name	Fluorescence spectroscopy, microscopy and applications
L-T-P-C	3-0-0-3
Prerequisites	NA
Intended for	MSc / M. Tech / PhD in Chemistry, Physics, nanotechnology and other discipline relevant to the course content
Distribution	Elective
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)

Course Contents

- **Introduction to fluorescence spectroscopy**

Light matter interaction, different processes when light absorbed by matter, light scattering, Fluorescence, Phosphorescence, absorption, Transition moments and transition probabilities, Einstein's coefficients, oscillator strength, Beer-Lambert law, polarizabilities, Frank Condon Principles. Steady state fluorescence, Jablonski diagram describing the excited processes, characteristics of fluorescence spectra, Kasha and Vavilov's rule, Stokes shift, radiative and nonradiative processes, overview of time and frequency domain measurement, fluorescence lifetime or decay, quantum yield and calculation, time correlated single photon counting (TCSPC), light source and electronics for TCSPC, Instrumentation of steady state and time resolved spectro photometer such as light source, detectors etc. [12 Lectures]

- **Solvent and environment effects**

Effect of solvent polarity on spectral shift, general solvent effect, The Lippert Mataga equation, specific solvent effect, temperature effect, additional factors affecting the spectral shift locally excited and internal charge transfer states, excited state intramolecular proton transfer, dynamics of solvent effect, time resolved emission spectra (TRES), picosecond spectral relaxation in solvent, theory for time dependent solvent relaxation, red edge excitation shift. [4 Lectures]

- **Quenching of fluorescence**

Reasons of fluorescence quenching, type of quenching, static and dynamic quenching, Stern Volmer equation, theory of static quenching, mechanism of dynamic quenching, theory of combined quenching, photo-induced electron transfer based quenching, application of quenching in proteins, DNA dynamics, sensors, molecular beacons based quenching, effect of temperature, viscosity on the quenching. [6 Lectures]

- **Fluorescence resonance Energy Transfer (FRET)**

Basics of energy transfer, donor acceptor pair, factors affecting the energy transfer, surface energy transfer (SET), difference between FRET and SET, comparison of quenching and FRET, distance dependent FRET, SET and quenching, metal enhanced fluorescence (MEF), mechanism of MEF, radiative decay engineering in MEF, surface plasmon coupled emission, energy transfer to multiple acceptors in one, two dimensions. [9 Lectures]

- **Fluorescence anisotropy**

Definition and theory of fluorescence anisotropy, relation of polarization to anisotropy, L and T format for anisotropy, effect of resonance energy transfer on anisotropy, effect of rotational diffusion on fluorescence anisotropy, time dependent anisotropy decay, rotational correlation time, applications of anisotropy on molecular interactions. [4 Lectures]

- **Fluorescence microscopy and single molecule detection**

Basic principles and applications of wide field fluorescence microscopy, fluorescence lifetime imaging microscopy (FLIM), confocal microscopy, laser scanning TCSPC FUM, single molecule detection (SMD), optical configuration of SMD, SMD detectors, single molecule based FRET, total internal reflection (TIRF), Fluorescence correlation spectroscopy (FCS), effect of concentration, diffusion coefficient, dual color fluorescence cross correlation (FCCS), applications of FCS and FCCS. [10 Lectures]

Text Books

1. Joseph R Lakowicz, **Principles of fluorescence spectroscopy**, 3rd Edition, Springer, 2010.
2. Peter Hinterdorfer, Antoine VanOijen, **Handbook of single molecule biophysics**, Springer, 2009.

References

1. R. Riglet, M. Orrit, T Basche, **Single Molecule Spectroscopy: Nobel Conference Lectures**, Springer Series in Chemical Physics, 2012.
2. Yves Engelborghs, Antonie Visser, **Fluorescence Spectroscopy and Microscopy**, Springer, 2014.
3. M Valeur and J. C. Brachon, **New Trends in Fluorescence Spectroscopy: Applications to Chemical and Life Sciences**, Springer, 2001.
4. Ewa M. Goldys, **Fluorescence Applications in Biotechnology and Life Sciences**, Wiley Blackwell, 2010.